INFOSYS VIRTUAL INTERNSHIP 6.0

FOOD TRENDS UNDERSTANDING CUSTOMER PREFERENCES

BATCH: 11 | GROUP: 1

TEAM: C

YEAR: 2026

Week -8

Submitted To:

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CONTENT :

1. Project Title

Infosys_FoodTrends: Understanding Customer Preferences

2. Project Objective

The primary objective of this project is to analyze restaurant data and derive insights that explain customer preferences and behavior. The project aims to understand how factors such as cuisine type, price range, online delivery, and customer engagement influence restaurant ratings and popularity. Another key objective is to present these insights in an interactive and visually appealing manner so that they can be easily interpreted by non-technical audiences, including media professionals and the general public.

Key objectives:

- Analyze customer preferences across different cuisines and regions
- Study customer engagement using ratings and vote counts
- Understand the impact of pricing on customer perception
- Communicate insights through interactive dashboards

3. Project Description

3.1 Overview

The Infosys Food Trends project was carried out as part of the Infosys Springboard Internship program and was executed as a team project involving five members. The dataset used for this project was sourced from Kaggle and contains detailed information about restaurants, including location, cuisine types, pricing, ratings, votes, and service availability. The project resulted in the development of seven interactive dashboards using Power BI, each designed to focus on a specific analytical dimension. Collectively, these dashboards provide a comprehensive understanding of food trends and customer preferences.

Overview highlights:

- Team-based project with five contributors
- Public dataset sourced from Kaggle
- Seven dashboards covering multiple analytical aspects
- Focus on real-world applicability

3.2 Project Approach

The project followed a well-defined and structured approach to ensure accuracy and quality. Initially, the dataset was collected from Kaggle and explored to understand its structure, attributes, and limitations. Data preprocessing was then carried out using Power Query to handle missing values, remove duplicates, and standardize categorical data. Additional calculated fields were created to support deeper analysis, such as price categories and engagement indicators. Once the data was cleaned and prepared, a data model was created in Power BI to establish proper relationships. The dashboard development phase involved selecting relevant KPIs, designing visuals, and adding slicers for interactivity. Throughout the project, dashboards were reviewed and refined through team discussions.

Approach summary:

- Dataset exploration and understanding
- Data cleaning and preprocessing
- Feature engineering and data modeling
- Dashboard creation and validation
- Continuous review and refinement

3.3 Technology Used

The project utilized Power BI as the primary visualization tool due to its ability to handle large datasets and create interactive dashboards. Power Query was used for data cleaning and transformation, ensuring the dataset was analysis-ready. DAX was applied to create calculated measures such as average ratings and vote-based metrics. Microsoft Excel was used for initial inspection and validation of the dataset before integration into Power BI.

Tools and technologies:

- Power BI for visualization and analytics
- Power Query for data preprocessing
- DAX for calculated measures
- Microsoft Excel for data validation

3.4 Insights from the Dashboard

The analysis generated meaningful insights into customer behavior and restaurant performance. Restaurants are largely concentrated in metropolitan areas, indicating strong urban demand. North Indian and Chinese cuisines dominate customer preferences, while online delivery services significantly increase customer engagement. Higher-priced restaurants generally receive better ratings, suggesting a link between price and perceived quality. Additionally, restaurants with more votes tend to maintain more stable ratings compared to those with fewer reviews.

Major insights:

- Urban regions dominate restaurant presence
- Certain cuisines show consistently higher demand
- Online delivery boosts engagement
- Higher pricing correlates with better ratings
- Vote count impacts rating stability

3.5 Real-World Impact for Media and Public Communication

The findings of this project have strong relevance in real-world scenarios, particularly for media and public communication. Media organizations can use the insights to publish data-driven articles on food trends and customer preferences. Consumers can leverage the dashboards to make informed dining decisions. Restaurant owners and business stakeholders can use the analysis to improve service offerings, pricing strategies, and customer engagement. Overall, the project demonstrates how data visualization can effectively communicate complex information to diverse audiences.

Real-world impact:

- Supports data-driven media reporting
- Enhances public awareness of food trends
- Aids business strategy and decision-making
- Demonstrates effective visual storytelling

4. Timeline Overview

The project was completed over a total duration of forty days. The execution was divided into milestone-based phases, with each milestone allocated one week. This structured timeline ensured systematic progress, effective task distribution, and timely completion of deliverables.

Timeline highlights:

Week	Activities Planned	Activities Completed
1	Dataset selection, understanding project requirements, and defining objectives	Selected Kaggle dataset, analyzed data structure, and finalized project objectives
2	Data cleaning, preprocessing, and transformation	Cleaned dataset using Power BI Power Query, handled missing values, standardized columns
3	Data modeling and creation of calculated measures	Established data relationships and created DAX measures for key metrics
4	Design of core dashboards and KPI visuals	Developed KPI cards, bar charts, and slicers for overall performance analysis
5	Customer and restaurant performance analysis	Built dashboards for customer behavior and restaurant-wise performance
6	Food category and trend analysis	Analyzed popular cuisines, items, and ordering trends using time-based visuals
7	Insight generation and dashboard optimization	Derived key insights, improved dashboard layout, and optimized visuals
8	Final review and documentation	Validated results, prepared project report, and finalized dashboards

5. Project Execution

5a. Key Milestones

The first milestone focused on data collection and preprocessing to prepare the dataset for analysis. During the second milestone, the first three dashboards were developed, focusing on market presence, cuisine demand, and customer engagement. The third milestone involved completing five dashboards and refining previously developed dashboards. The final milestone was dedicated to completing all seven dashboards, validating insights, and optimizing performance.

Milestone Overview:

Milestones	Description	Date
Milestone 1	Data Preparation & Transformation: Collected dataset from Kaggle, cleaned and transformed data using Power BI Power Query, and created calculated columns and measures.	16-01-2026
Milestone 2	Core Dashboard Development: Designed interactive dashboards to analyze business performance, customer behavior, and restaurant performance using KPIs and trend visuals.	23-01-2026
Milestone 3	Food & Category Trend Analysis: Analyzed food categories, popular items, ordering patterns, and rating trends through category-wise visualizations.	30-01-2026
Milestone 4	Strategic Insights & Optimization: Derived actionable insights on restaurant consistency and customer satisfaction, optimized dashboards, and completed project documentation.	06-02-2026

Milestone summary:

Milestone 1: Data Preparation & Transformation

The first milestone focused on collecting and preparing the dataset for analysis. The dataset was sourced from Kaggle and contained restaurant, customer, order, and rating-related information. An initial exploration of the dataset was conducted to understand its structure, attributes, and data types.

Data cleaning was performed using **Power BI Power Query**, where missing values, inconsistent entries, and redundant records were identified and handled appropriately. Columns were standardized to maintain uniformity in formats such as categorical values, dates, and numerical fields. Unnecessary columns were removed to improve data model efficiency.

Additionally, calculated columns and DAX measures were created to support analytical requirements such as total revenue, order count, average ratings, and customer-related metrics. This milestone ensured that the dataset was accurate, consistent, and analysis-ready, forming a strong foundation for subsequent dashboard development.

Milestone 2: Core Dashboard Development

In the second milestone, the focus shifted to designing and developing the core dashboards in Power BI. A structured dashboard layout was planned to represent overall business performance, customer behavior, and restaurant performance in a clear and interactive manner.

Key visual elements such as **KPI cards, bar charts, line charts, and slicers** were implemented to present high-level metrics including revenue, total orders, customer count, and average ratings. Relationships between tables were verified to ensure accurate filtering and cross-visual interactions.

The dashboards were designed with usability in mind, enabling users to quickly interpret trends and compare performance across cities, restaurants, and time periods. This milestone resulted in functional dashboards that provided a comprehensive overview of the dataset.

Milestone 3: Food & Category Trend Analysis

The third milestone involved detailed analysis of food categories, popular items, and ordering patterns. Category-wise performance was examined to identify the most preferred cuisines and food items based on order frequency and customer ratings.

Trend analysis was conducted to study ordering behavior across different time periods, days, and customer segments. Visualizations were created to highlight variations in demand, peak ordering periods, and customer preferences. Rating trends were also analyzed to understand the relationship between food quality, consistency, and customer satisfaction.

This milestone helped uncover deeper insights into customer behavior and food trends, enabling a more granular understanding of what drives demand and performance in the restaurant ecosystem.

Milestone 4: Strategic Insights & Optimization

The final milestone focused on generating actionable insights and optimizing the dashboards for performance and clarity. Key insights related to restaurant consistency, quality stability, customer retention, and revenue concentration were derived from the analytical findings.

Dashboards were refined by optimizing visuals, reducing clutter, and improving interactivity through effective use of slicers and filters. Measures and calculations were validated to ensure accuracy and reliability of insights.

Finally, project documentation was prepared, summarizing the methodology, dashboards, challenges, and outcomes. This milestone ensured that the project delivered meaningful business insights and a polished analytical solution ready for evaluation and future enhancement.

5b. Project Execution Details

Step 1: Data Collection & Finalization

- The dataset was sourced from Kaggle (Global Restaurant Dataset).
- The dataset was reviewed and finalized collectively.
- Project objectives and key business questions were defined.

Step 2: Data Cleaning & Preparation

- Data was imported into Power BI.
- Cleaning was performed using Power Query Editor:
    - Removal of unnecessary columns
    - Handling missing values
    - Data type standardization

Step 3: Data Modeling & DAX Implementation

- Calculated columns were created using **DAX**, including:
    - *Vote* Bucket (engagement grouping)
    - *Quality* Category (rating reliability)
    - *Action* Category (business prioritization)
- Measures were developed for votes, ratings, and restaurant counts.

Step 4: Dashboard Development

- Dashboards were created milestone-wise:
    - Market & Cuisine Analysis
    - Pricing & Service Impact
    - Quality Stability & Strategic Actions
- Visuals were selected based on analytical relevance and storytelling.

Step 5: Review & Refinement

- Regular team discussions ensured alignment and accuracy.
- Insights were validated and refined.
- Mentor feedback was incorporated iteratively.

Step 6: Documentation & Repository Management

- Final dashboards were integrated.
- Report and presentations were prepared.
- GitHub repository and documentation were finalized.

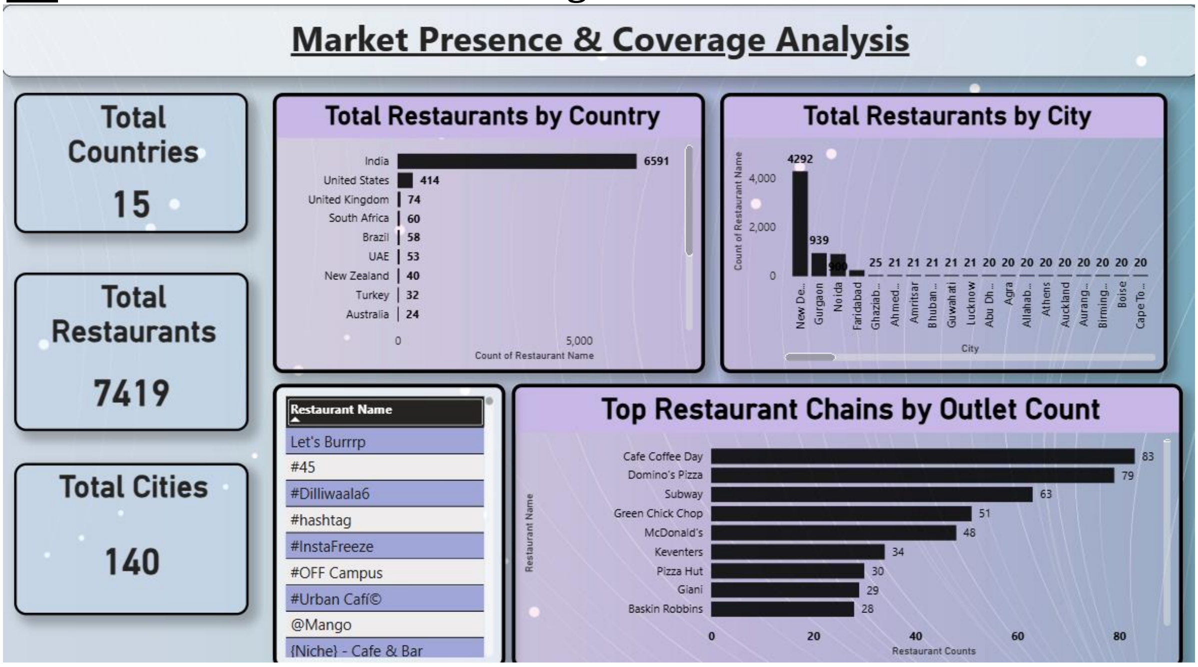
Execution Highlights:

- Strong team-based collaboration across all project stages
- Equal distribution and shared ownership of responsibilities
- Continuous review and iterative improvement process
- Consistent dashboard design and accurate analytical validation

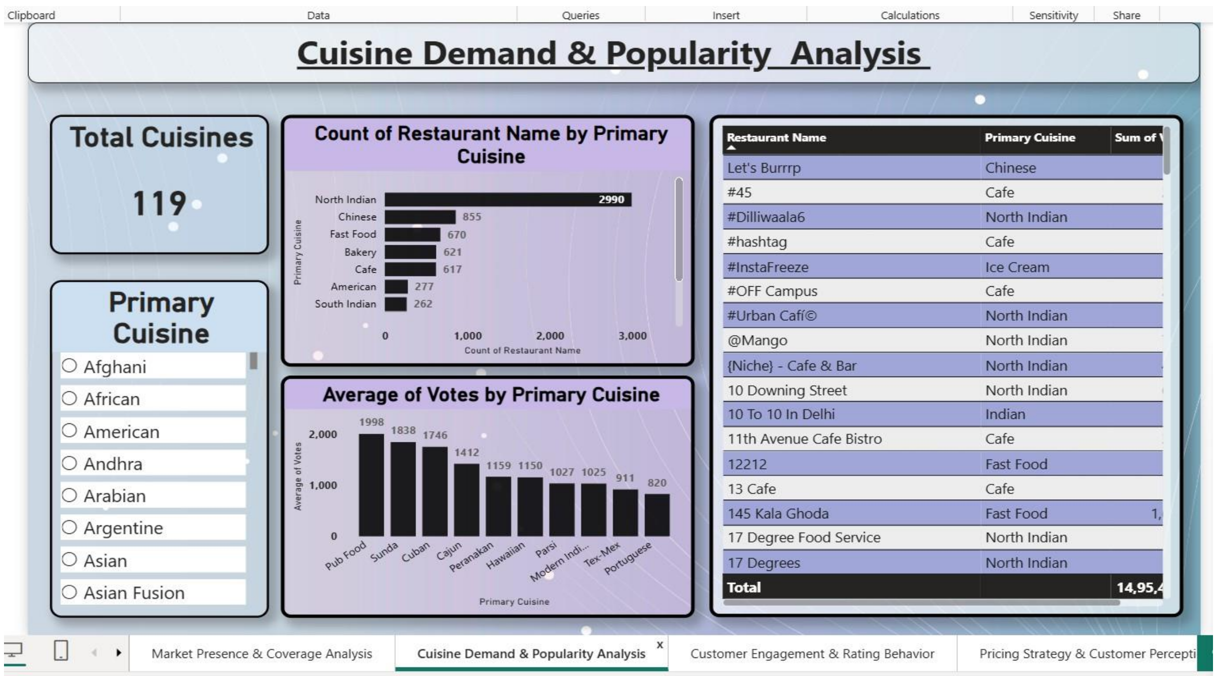
6. Snapshots / Screenshots

This section contains screenshots of all seven dashboards developed during the project. The screenshots visually demonstrate the KPIs, filters, and insights derived from the analysis and support the findings discussed in this report. Included dashboards are:

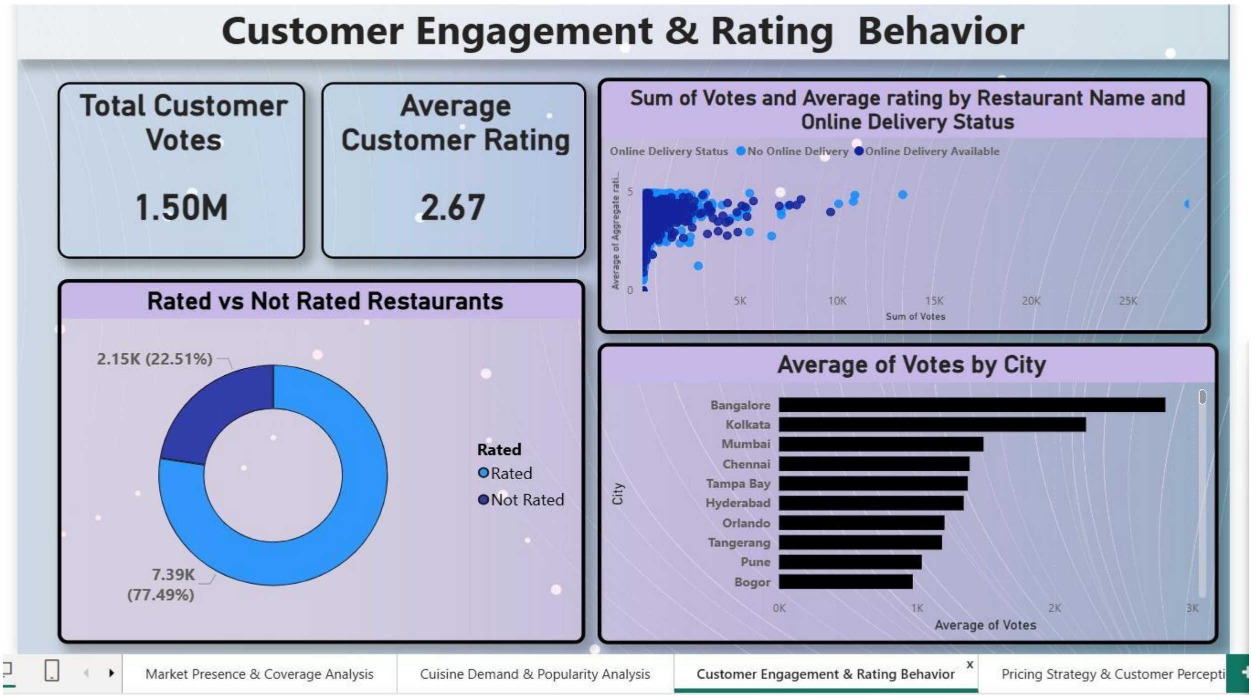
6.1 Market Presence & Coverage



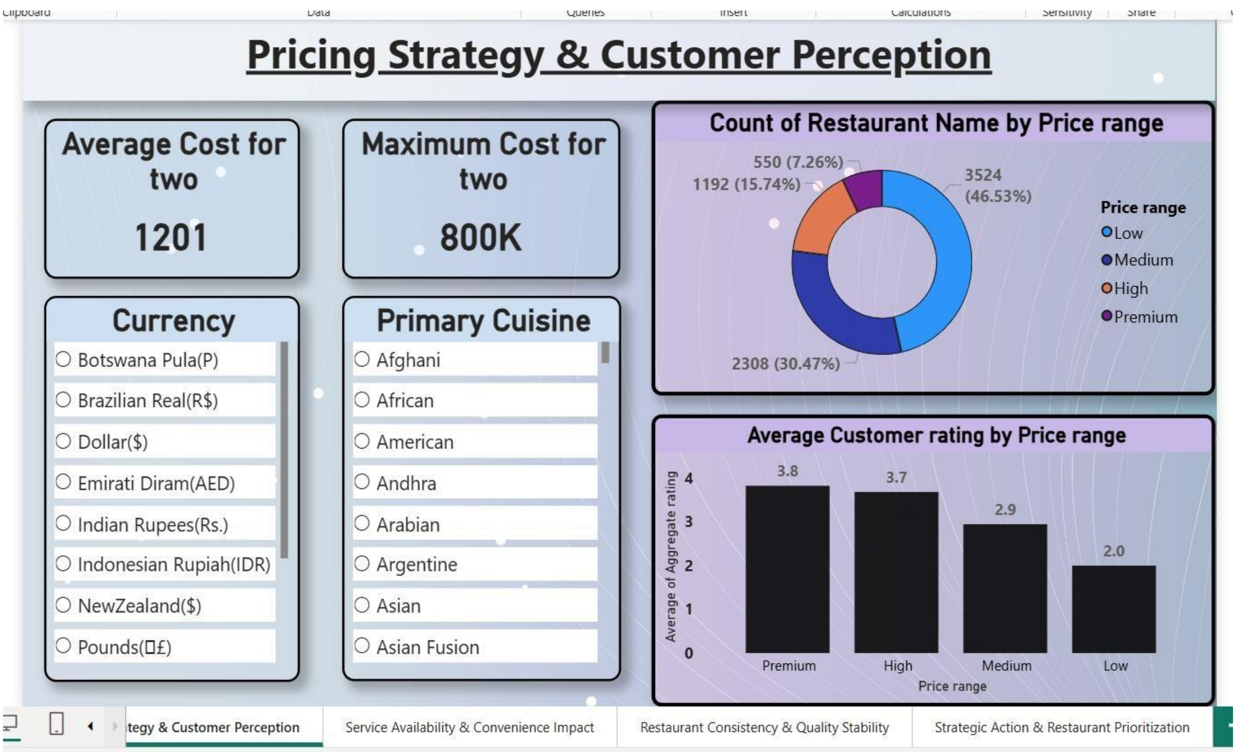
6.2 Cuisine Demand & Popularity



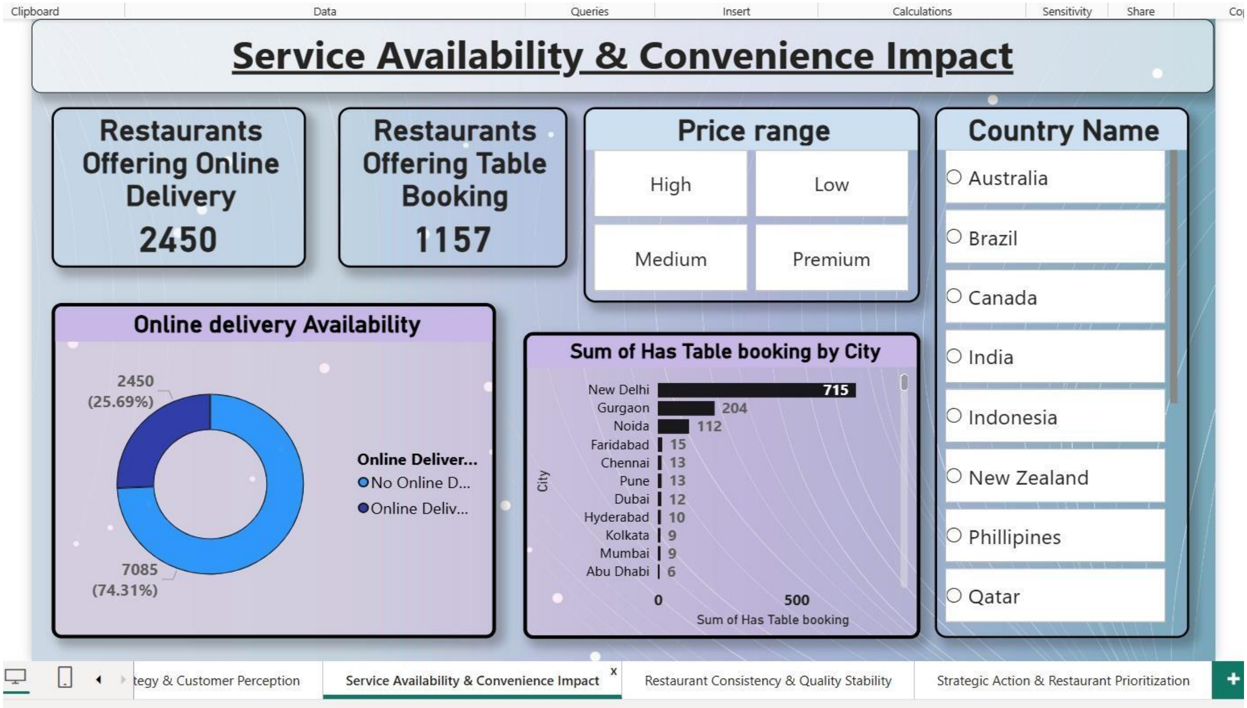
6.3 Customer Engagement & Ratings



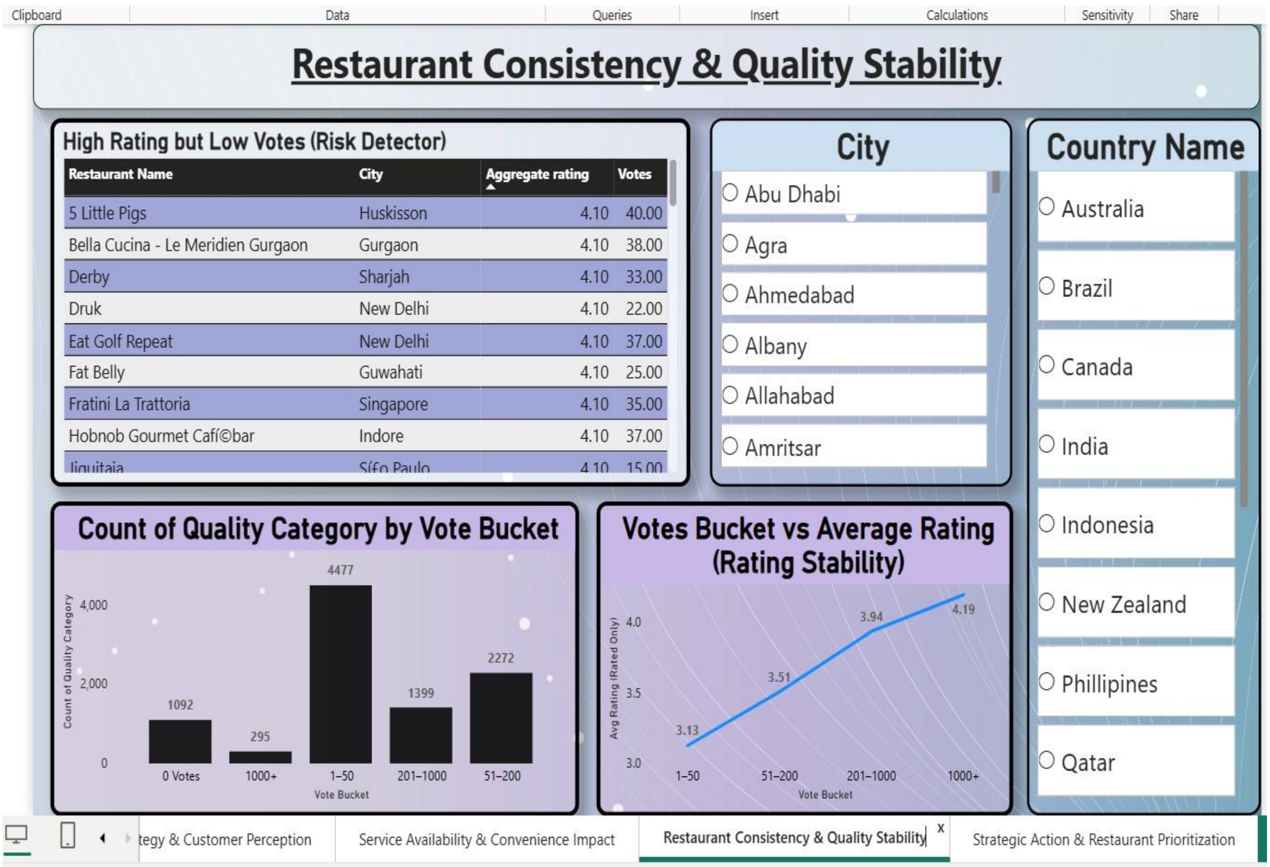
6.4 Pricing Strategy & Perception



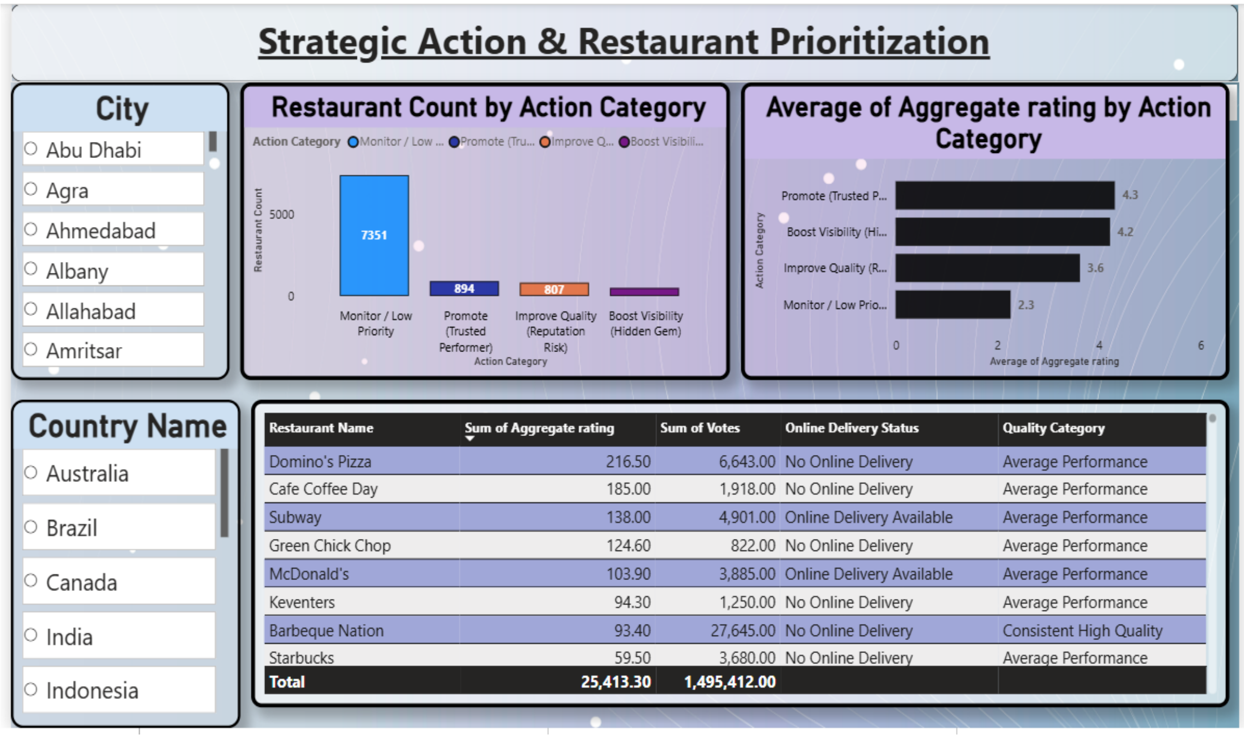
6.5 Service Availability Impact



6.6 Quality Stability Analysis



6.7 Strategic Restaurant Categorization



Restaurant Name	Sum of Aggregate rating	Sum of Votes	Online Delivery Status	Quality Category
Domino's Pizza	216.50	6,643.00	No Online Delivery	Average Performance
Cafe Coffee Day	185.00	1,918.00	No Online Delivery	Average Performance
Subway	138.00	4,901.00	Online Delivery Available	Average Performance
Green Chick Chop	124.60	822.00	No Online Delivery	Average Performance
McDonald's	103.90	3,885.00	Online Delivery Available	Average Performance
Keventers	94.30	1,250.00	No Online Delivery	Average Performance
Barbeque Nation	93.40	27,645.00	No Online Delivery	Consistent High Quality
Starbucks	59.50	3,680.00	No Online Delivery	Average Performance
Total	**25,413.30**	**1,495,412.00**		

7. Challenges Faced

During the development of the project, several technical and analytical challenges were encountered. These included handling missing and inconsistent data entries, managing relatively large datasets within Power BI, selecting meaningful performance indicators, and interpreting rating trends where vote counts were low. Each of these challenges required careful analysis and structured problem-solving to ensure the reliability and clarity of the final dashboards.

Key Challenges and Resolutions:

- **Challenge 1: Inconsistent and Missing Data**

  Resolution: Applied systematic data cleaning and preprocessing techniques using Power Query. Missing values were handled appropriately, inconsistent entries were standardized, and unnecessary columns were removed to improve data quality and consistency.

- **Challenge 2: Identifying Meaningful KPIs**

  Resolution: Carefully analyzed business requirements to determine relevant metrics such as total revenue, order frequency, average ratings, and consistency scores. DAX measures were created to accurately calculate and represent these KPIs.

- **Challenge 3: Dashboard Clutter and Readability Issues**

  Resolution: Improved dashboard layout by reducing redundant visuals, aligning elements properly, and incorporating slicers for dynamic filtering. This enhanced user experience and ensured better interpretability of insights.

- **Challenge 4: Performance Optimization in Power BI**

  Resolution: Optimized DAX calculations, minimized redundant columns, and improved data model relationships to enhance dashboard performance and reduce loading time.

8. Learnings & Skills Acquired

8.1 Technical Skills

This project enhanced practical knowledge of Power BI, including dashboard creation, data visualization, and performance optimization. Hands-on experience was gained in Power Query for data cleaning and transformation, as well as in developing DAX measures and calculated columns. It also improved understanding of data modeling and building interactive dashboards.

8.2 Analytical and Problem-Solving Skills

The project strengthened analytical thinking by identifying meaningful KPIs and interpreting business trends from raw data. It improved the ability to analyze patterns, validate insights, and solve data-related challenges effectively.

8.3 Soft Skills and Team Collaboration

Working on this project improved communication, presentation, and time management skills. It also encouraged structured planning and collaborative discussion for refining dashboards and insights.

8.4 Domain Knowledge and Application

The project provided practical exposure to food industry analytics, including customer behavior analysis, revenue trends, cuisine preferences, and the importance of quality consistency in restaurant performance.

9. Testimonials from Team

Piyush Nitnaware:
Contributed to dataset collection, initial data understanding, and data cleaning using Power Query. Assisted in identifying missing values and standardizing columns for accurate analysis.

Vaishnavi Shah:
Worked on data transformation and creation of calculated columns and DAX measures. Contributed to defining key performance indicators such as revenue, order count, and average ratings.

Sam Joseph:
Led the development of core dashboards, including KPI visuals, bar charts, and trend analysis charts. Ensured proper data modeling within Power BI.

Raghav Vyas:
Focused on food category analysis, customer behavior insights, and restaurant performance comparison. Helped interpret rating trends and identify meaningful business insights.

Harini M:
Handled dashboard optimization, layout refinement, and report documentation. Contributed to insight validation, performance improvement, and final presentation preparation.

Team Reflection:
All team members expressed that the project provided valuable hands-on experience in data analytics and strengthened collaborative learning. The equal distribution of responsibilities, continuous peer discussions, and iterative review process were highlighted as key strengths that contributed to the successful completion of the project.

10. Conclusion

The *Food Trend Analysis and Restaurant Performance Dashboard* project demonstrates the effective application of data analytics and business intelligence techniques using Power BI. The primary objective was to transform raw restaurant and customer data into meaningful, actionable insights to support data-driven decision-making. This objective was successfully achieved through systematic data preparation, modeling, and interactive visualization.

The project involved thorough data cleaning and preprocessing using Power Query to ensure data accuracy and consistency. Calculated columns and DAX measures were developed to derive key metrics such as total revenue, order volume, average ratings, and customer demographics, enabling reliable and structured analysis.

The dashboards revealed important business insights, including revenue concentration in major cities such as Delhi (₹105M+), peak ordering on Fridays (34K+ orders), and an average customer age of 24 years. Cuisine analysis highlighted strong demand for North Indian and Chinese categories, while rating trends emphasized the importance of quality consistency for customer retention. Growth patterns indicated early expansion followed by market stabilization.

Overall, the project enhanced technical skills in data transformation, DAX, and dashboard design while strengthening analytical thinking and business interpretation abilities. It effectively showcases how Power BI can convert complex datasets into interactive dashboards that support strategic planning and performance evaluation.

11. Acknowledgements

We would like to express our heartfelt gratitude to:

Infosys Springboard Team – for organizing this valuable internship opportunity and providing a structured learning platform.

Our Mentor – for her continuous guidance, constructive feedback, and encouragement throughout the project.

My Teammates – for their collaboration, support, idea sharing, and teamwork during all phases of the project.

We also sincerely appreciate the opportunity to work on a real-world dataset that allowed us to apply analytical concepts in a practical environment. The exposure to Power BI, data modeling, and structured project execution has significantly enhanced our technical knowledge and problem-solving skills.

The collaborative environment, regular reviews, and milestone-based execution helped us develop not only technical expertise but also essential skills such as communication, accountability, time management, and professional coordination.

This internship has been a significant step in our professional growth and technical journey.